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30 September 1964
SUPERSEDING
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18 August 1961

MILITARY SPECIFICATION

VOICEMITTER-OUTLET VALVE ASSEMBLY, C29R1

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of protective mask voicemitter-outlet valve assembly.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129 - Marking for Shipment and Storage.

MIL-STD-282 - Filter Units, Protective Clothing, Gas-Mask Components and Related Products: Performance Test Methods.

FSC 4240

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DRAWINGS

EDGEWOOD ARSENAL

DL5-81-3 - Voicemitter-Outlet Valve, C29R1.

(All drawings referenced in this specification are listed on Drawing DL5-81-3.)

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification Ratings, Rules and Regulations.

(Application for copies of these uniform classification rules should be addressed to the Uniform Classification Committee, 202 Union Station, Chicago 6, Illinois.)

3. REQUIREMENTS

3.1 Materials and components. All materials and components shall conform to the specifications and drawings forming a part of this specification.

3.2 Manufacture and assembly. The voicemitter-outlet valve shall be manufactured and assembled as specified on Drawing D5-81-3 and subsidiary drawings listed thereon.

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3.3 Diaphragm assembly.

3.3.1 DOP smoke penetration. The diaphragm assembly shall show a DOP smoke penetration of no more than 0.001 percent when tested as specified in 4.7.1.

3.3.2 Diaphragm tension. Tension on the diaphragm shall be such that the center of the diaphragm shall be deflected no less than 0.025 inch nor more than 0.040 inch, without seating on the baffle, when tested as specified in 4.7.2.

3.3.3 GB penetration of diaphragm material. The diaphragm material shall resist the penetration of liquid GB for no less than two hours when tested as specified in 4.7.3. The initial 15 linear feet of each roll of diaphragm material used for preproduction or regular production shall be submitted to the Government for GB testing.

3.4 Resistance to airflow. The assembled outlet valve shall have a resistance not exceeding a pressure differential of 22 millimeters of water at an airflow rate of 85 liters per minute when tested as specified in 4.7.4.

3.5 Outlet valve leakage. Leakage through the assembled outlet valve shall not exceed 15 milliliters of air per minute, at a pressure differential of 25 millimeters of water, when tested as specified in 4.7.5.

3.6 DOP smoke penetration. DOP smoke penetration of the voicemitter-outlet valve assembly with the opening in the valve seat sealed shall not exceed 0.001 percent when tested as specified in 4.7.6.

3.7 Preproduction sample. Prior to the start of regular production,

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a sample of 10 voicemitter-outlet valve assemblies shall be produced using the same methods, materials, and equipment as will be used during regular production. Any change in materials or operating conditions shall require a new preproduction sample.

3.8 Workmanship. The voicemitter-outlet valve assembly shall be free from foreign matter, and damage such as distortion, dents, roughened valve seat, and torn disc that would impair its function.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Supplier's responsibility. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to insure that supplies and services conform to specified requirements.

4.1.2 Government's responsibility. The Government will be responsible for the performance of the testing in 4.7.3. Sample diaphragm material shall be forwarded to the laboratory designated by the contracting officer.

4.2 Special provisions.

4.2.1 Alternative inspection. The supplier may utilize any alternative inspection procedure which will provide equal or better assurance of quality by submitting a written proposal with justification and obtaining written approval from the Government prior to instituting

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the procedure. In case of dispute, the procedures of this specification will govern.

4.2.2 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2.3 Inspection substation. If the supplier elects to establish inspection substations for the purpose of conducting inspection for certain listed defects, the Acceptable Quality Levels (AQL's) for each inspection substation shall be equal to the AQL which is the next numerically lower value than the AQL specified for the total listed defects. The requirements of MIL-STD-105 are applicable to each substation and to the end item.

4.3 Preproduction sample inspection.

4.3.1 Examination. The sample voicemitter-outlet valve assemblies shall be inspected for all requirements of the drawings and this specification.

4.3.2 Tests. The sample voicemitter-outlet valve assemblies shall be tested in accordance with paragraph 4.7.

4.3.3 Acceptance/rejection criteria. The preproduction sample shall meet the examinations and tests specified by 4.3.1 and 4.3.2 to be acceptable. The supplier shall obtain written approval from the contracting officer before proceeding with regular production.

4.4 Lotting. A lot shall consist of the voicemitter-outlet valve assemblies produced by one manufacturer, at one plant, from the same materials, and under the same manufacturing conditions. No more than

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one lot of outlet valve discs shall be represented in any one lot of voicemitter-outlet valve assemblies.

4.5 Sampling. Sampling shall be conducted in accordance with MIL-STD-105.

4.6 Inspection provisions.

4.6.1 For examination and nondestructive tests. Sample voicemitter-outlet valve assemblies shall be examined in accordance with the classification of defects and with MIL-STD-105.

4.6.2 For critical defects. Each item in the lot shall be inspected for critical characteristics listed in the classification of defects.

4.6.3 Classification of defects.

4.6.3.1 Diaphragm assembly (Dwg C5-81-13).

Categories	Defects	Acceptance standards
Critical:		
1	DOP smoke penetration	4.7.1
Major:	AQL 1.0 percent defective	
101	Components missing or incorrectly assembled	
102	Outside diameter (2.364) incorrect	
103	Outside periphery thickness (0.130) incorrect	
104	Workmanship (see 3.8)	
Major A:	AQL 0.25 percent defective	
150	Diaphragm tension	4.7.2

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4.6.3.2 Voicemitter-outlet valve, C29R1, assembly (Dwg D5-81-3).

Categories	Defects	Acceptance standards
Critical:		
1	Outlet valve leakage	4.7.5
2	DOP smoke penetration	4.7.6
Major:	AQL 1.0 percent defective	
101	Component missing, incorrect, or incorrectly assembled	
102	Sealing ring loose	
103	Workmanship (see 3.8)	
Major A:	AQL 0.25 percent defective	
150	Outlet valve resistance to airflow	4.7.4
Minor:	AQL 2.5 percent defective	
201	Protective finish missing	

4.7 Tests. Tests shall be conducted as follows:

4.7.1 DOP smoke penetration. Mount the diaphragm assembly in a suitable fixture, with an airtight seal around the periphery, and spray DOP smoke over the front side of the assembly. The DOP smoke shall have a concentration of 100 to 150 micrograms per liter and a maximum particle size of 0.4 micron diameter. Maintain a negative pressure of 150 millimeters of water on the back side of the assembly and, with a flow rate of 16 liters per minute, measure the DOP smoke penetration through the assembly by means of a suitable smoke density measuring device.

4.7.2 Diaphragm tension. Determine the diaphragm tension using Edgewood Arsenal Q41 Tension Tester. Apply 114 ± 1 grams of weight on the center of the diaphragm by means of a rod which passes freely

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through the center hole of the cupped baffle. The total weight of the rod, with attached linkage, shall be 19 ± 1 gram. Measure only the deflection of the diaphragm caused by the addition of the 114 gram weight, using a deflection gage accurate to ± 0.001 inch. Use an additional 1 to 5 gram weight to determine that the diaphragm has not seated or bottomed on the baffle.

4.7.3 GB penetration. Determine the resistance of the diaphragm material to GB penetration in accordance with the procedure specified in method 206.1.1 of MIL-STD-282.

4.7.4 Resistance to airflow. Test the outlet valve assembly of the voicemitter for resistance to airflow using Edgewood Arsenal Q104 Inlet and Outlet Valve Resistance Indicator.

4.7.5 Outlet valve leakage. Test voicemitter-outlet valve assembly for leakage using Edgewood Arsenal M4A1 Outlet Valve Leakage Indicator.

4.7.6 DOP smoke penetration. Mount the assembly in a suitable fixture that provides an airtight seal around the periphery. The opening of the voicemitter outlet valve shall be sealed off without sealing off the valve seat and frame joint. Expose the front side of the assembly to a concentration of 100 to 150 micrograms per liter of DOP smoke of 0.4 micron maximum particle size. Maintain a negative pressure of 150 millimeters of water on the back side of the assembly, and with a flow rate of 16 liters per minute, measure the DOP smoke penetration through the assembly by means of a suitable smoke density measuring device.

5. PREPARATION FOR DELIVERY (For repair parts packaging see 6.3)

5.1 Packaging, level C. Voicemitter-outlet assemblies shall be packaged to afford adequate protection from the supply source to the

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first receiving activity for immediate use.

5.2 Packing, level C. Voicemitter-outlet valve assemblies, packaged as specified in 5.1, shall be packed in such manner as to insure carrier acceptance and safe delivery to first domestic destination. Containers shall comply with Uniform Freight Classification rules, regulations of other carriers applicable to the mode of transportation.

5.3 Marking. In addition to any special marking required by the contract or order, unit packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The item covered by this specification is a combination voicemitter and outlet valve, intended for use with the ABC-M17 protective mask.

6.2 Ordering data. Procurement documents should specify the title, number, and date of this specification.

6.3 Repair parts packaging. When repair parts packaging is required for items authorized for stocking as repair parts, the repair parts packaging data sheets covering the voicemitter-outlet valve assembly should be used. (Copies of repair parts packaging data sheets may be obtained from the Director for Engineering and Industrial Services, ATTN: Chief, Design Branch, US Army Edgewood Arsenal, Edgewood Arsenal, Maryland - 21010.)

6.4 Component specifications. Following are listed the component specifications for the C29R1 voicemitter-outlet valve:

Discs, Outlet Valve - MIL-D-10132

Voicemitter-Outlet Valve, C29R1, Components for - MIL-V-50067

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6.5 International interest. Certain provisions of this specification are the subject of ABC international standardization agreement. When amendment, revision, or cancellation of this specification is proposed, the departmental custodians will inform their respective Departmental Standardization Offices so that appropriate action may be taken respecting the international agreement concerned.

Custodians:

Army - MU

Navy - SH

Air Force - 84

International interest (see 6.5)

Review activities:

Army - MD

Navy - SH

User activities:

Navy - MC

Air Force - 84

Preparing activity:

Army - MU(EA)

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